

Cancer Patterns in Hmong in Minnesota, 2000 to 2012

Mosunmoluwa Oyenuga, MD ¹; John K. Yang, MPH¹; Anna E. Prizment, PhD, MPH^{1,2}; Sally Bushhouse, DVM, MPH, PhD³; Ellen W. Demerath, PhD¹; and Logan G. Spector, PhD^{2,4}

BACKGROUND: Minnesota has the second largest Hmong population in the United States. The objective of the current study was to estimate the cancer incidence among Hmong individuals in Minnesota between 2000 and 2012 to determine targets for screening and interventions. **METHODS:** Cancer cases in Minnesota between 2000 and 2012 were obtained from the Minnesota Cancer Surveillance System, and proportional incidence ratios (PIRs) were calculated. The 2000 and 2010 US Census reports were used to obtain total population estimates. Age-adjusted cancer incidence rates (AAR) and 95% confidence intervals (95% CIs) were calculated for Hmong individuals, Asian/Pacific Islander individuals, and all Minnesotans using direct method and Poisson regression. **RESULTS:** Compared with all Minnesotans, the Hmong had elevated PIRs and AARs for malignancies related to infections, including nasopharyngeal, stomach, liver, and cervical cancers. The AAR ratios in Hmong versus all Minnesotans were found to be significantly increased for nasopharyngeal (AAR, 15.90; 95% CI, 9.48-26.68), stomach (AAR, 2.99; 95% CI, 2.06-4.33), liver (AAR, 1.77; 95% CI, 1.04-3.02), and cervical (AAR, 3.88; 95% CI, 2.61-5.77) cancers. The AARs in Hmong versus all Minnesotans were significantly lower for all-cause cancer (AAR, 0.39; 95% CI, 0.35-0.44); cancers of the breast, lung, and colorectum; melanoma; and non-Hodgkin lymphoma. Compared with Asian/Pacific Islander individuals, the rates in Hmong were significantly higher for melanoma and cervical cancer, with AAR ratios of 2.23 (95% CI, 1.09-4.56) and 1.59 (95% CI, 1.01-2.49), respectively. **CONCLUSIONS:** Compared with all Minnesotans, the Hmong have an increased incidence of cancers related to infectious agents. These findings indicate a need for cancer prevention and screening programs in this population. *Cancer* 2018;124:3560-66. © 2018 American Cancer Society.

KEYWORDS: age-adjusted cancer incidence rates (AARs), Asian/Pacific Islander (API), cancer, Hmong, Minnesota.

INTRODUCTION

The Hmong began immigrating to the United States from Laos and Thailand in the 1970s as refugees of the Vietnam War, and many settled in Minnesota. Between 2004 and 2010, Minnesota received 5297 Hmong immigrants.¹ According to the 2010 US Census, Minnesota currently is the state with the second largest Hmong population (66,181) after California (91,224).² In 2010, the Hmong constituted 1.2% of all Minnesotans and 27% of all Asian/Pacific Islander (API) residents in the state,² and the number of Hmong in Minnesota is rapidly increasing. The Hmong population in Minnesota increased by 52.6%, the API population by 5.8%, and the overall Minnesota population by 7.8% between 2000 and 2010.^{3,4} The higher increase in the Hmong population may be due to the continuing immigration and large average family size. In 2010, the average Minnesota Hmong family size was approximately 5.59 with a median age of 20.3 years, whereas the average family size in all of Minnesota was only 3.05, with a median age of 37.4 years.⁴ However, the marked percentage increase in the Hmong can be partially attributed to the small size of the Hmong population, so that a smaller denominator resulted in a large percentage increase. The establishment of the Hmong population in the United States draws attention to their health needs. Conditions that continue to challenge this group include tuberculosis, hepatitis B virus (HBV), diabetes, cardiovascular disease, and cancer.⁵

In a single study conducted in Minnesota between 1988 and 1999, the Hmong had elevated proportional incidence ratios (PIRs; the incidence of a specific cancer compared with the incidence of total cancer) for stomach, liver, cervical, and nasopharyngeal cancers compared with the general population of Minnesota.⁶ The incidence of nasopharyngeal cancer was especially high, with a PIR of 39.39 (95% confidence interval [95% CI], 21.01-66.86), whereas the PIRs for several other cancers were lower in the Hmong, with the lowest PIR observed for prostate cancer (PIR, 0.16; 95% CI, 0.03-0.44). In parallel to the Minnesota study, a study performed in California in 2005 found that the Hmong had a lower age-adjusted cancer incidence rate (AAR) of all-cause cancer, but higher AARs for liver, stomach, cervical, and

Corresponding author: Anna E. Prizment, PhD, Division of Epidemiology and Community Health, University of Minnesota School of Public Health, 1300 S 2nd St, Suite 300, Minneapolis, MN 55454; prizm001@umn.edu

¹Division of Epidemiology and Community Health, University of Minnesota School of Public Health, Minneapolis, Minnesota; ²Masonic Cancer Center, University of Minnesota, Minneapolis, Minnesota; ³Minnesota Cancer Reporting System, Minnesota Department of Health, St. Paul, Minnesota; ⁴Division of Pediatric Epidemiology and Clinical Research, Department of Pediatrics, University of Minnesota, Minneapolis, Minnesota

The first 2 authors contributed equally to this article.

DOI: 10.1002/cncr.31592, **Received:** January 23, 2018; **Revised:** May 11, 2018; **Accepted:** May 16, 2018, **Published online** July 5, 2018 in Wiley Online Library (wileyonlinelibrary.com)

nasopharyngeal cancers, as well as leukemia and non-Hodgkin lymphoma compared with non-Hispanic white individuals.⁷ Thus, higher rates of cancers related to infectious agents are noted in the Hmong population. The incidence of these cancers historically has been lower in the United States compared with East Asia, and vaccination programs in the United States have reduced further the spread of human papillomavirus (HPV) as a cause of cervical cancer and HBV as a cause of hepatocellular cancer. Therefore, targeting public health interventions toward the Hmong may help to reduce the risk of these cancers in this population.

Our long-term goal is to decrease the risk of cancer development in the Hmong living in Minnesota. To develop effective public health interventions, it is necessary as a first step to determine cancer patterns and their changes over time in this population. Hence, the primary objective of the current study was to estimate Hmong PIRs between 2000 and 2012 and compare them with the PIRs reported for 1988 through 1999 in a previous study by Ross et al.⁶ The second objective was to estimate for what to the best of our knowledge is the first time AARs for all-cause and specific cancers and compare them with the rates in all Minnesotans and Minnesotans of Asian/Pacific Islander (API) ancestry.

MATERIALS AND METHODS

The Minnesota Cancer Surveillance System (MCSS) database was used to identify cancers in the entire Minnesota population and in Hmong in Minnesota between 2000 and 2012. The MCSS is a population-based cancer registry for the state, and has been in operation since 1988. Cancers diagnosed in Minnesota residents are identified in 2 ways: 1) submission of pathology reports carrying a diagnosis of cancer by pathology laboratories; and 2) submission of standardized case abstracts by hospital-based cancer registries for each patient with cancer diagnosed or treated at the facility. An external audit of the MCSS performed in March 2008 by the National Program of Cancer Registries estimated that case completeness was approximately 99.7% and data accuracy was approximately 96.5%.⁸ The MCSS is funded by a state appropriation and a cooperative agreement with the Centers for Disease Control and Prevention through the National Program of Cancer Registries.

Hmong and API cancer cases diagnosed between 2000 and 2012 were identified through the MCSS race reporting. Information regarding race in the MCSS was obtained from health care facilities that recorded patients' race and was supplemented using manual examination by MCSS staff of the records in those facilities and birth and

death records when a case was reported without information regarding race/ethnicity. The denominators for analysis came from the US Census Bureau, which obtained self-reported race information in response to the 2000 or 2010 census questionnaire.^{2,3} Census information was obtained through a search for "Hmong" and "Minnesota" using the American FactFinder.³

Data Analysis

We calculated age-adjusted and sex-adjusted PIRs as described by Ross et al.⁶ Observed-to-expected ratios were calculated. The observed number is the actual number of Hmong cancer cases for a particular cancer site. In generating the expected number, cancer cases were stratified into 20-year age groups (ages birth to 19 years, 20-39 years, 40-59 years, 60-79 years, and ≥ 80 years) and by sex. The expected number in each stratum was the number of site-specific cancers in all Minnesotans divided by the number of total cancers in all Minnesotans, multiplied by the number of total cancers in the Hmong. The total number was found by summing the expected values from each stratum. The 95% CIs were calculated using the methods described by Ury and Wiggins.⁹

We also calculated average annual AARs for all-cause cancer and site-specific cancers for Hmong individuals in Minnesota, API individuals in Minnesota, and all Minnesotans using the direct method,¹⁰ and these were verified using Poisson regression. The numerator included all incident cancer cases for the corresponding group diagnosed between 2000 and 2012. The denominator was calculated as the average of US Census population counts in 2000 and 2012 multiplied by 13. The 2000 US population was used as the standard weight. Adjustment was performed through 5-year age groups starting from birth to 4 years and ending at age ≥ 80 years. In addition, Poisson regression was used to calculate AAR ratios and 95% CIs for all groups: Minnesota Hmong, Minnesota API, and all Minnesotans.

Microsoft Excel (version 14.4.1; Microsoft Corporation, Redmond, Washington) was used to calculate PIRs and AARs. SAS statistical software (version 9.4; SAS Institute Inc, Cary, North Carolina) was used to verify the AARs and calculate the rate ratios, 95% CIs, and *P* values.

RESULTS

Between 2000 and 2012, there were 443 incident cancer cases in the Hmong population reported and ascertained by the MCSS: 197 cancer cases in males and 246 cancer cases in females. The percentage of male cancer cases was similar in the Hmong (44%) and API (45%), and lower than in all Minnesotans (52%). The percentage of female

cancer cases was similar among the Hmong (56%) and API (55%), and was higher than in all Minnesotans (48%). The average age of patients with cancer among Hmong was 53.5 years, and was 55.8 years in API patients and 64.9 years in all Minnesotans (Table 1).

The PIRs for cancers diagnosed between 2000 and 2012 are shown in Table 2.⁶ The PIRs for all cancers except non-Hodgkin lymphoma (NHL) were not significantly different when comparing these 2 time periods. The PIR for NHL was significantly higher in 2000 through 2012 (PIR, 1.83; 95% CI, 1.30-2.50) compared with 1988 through 1999 (PIR, 0.57; 95% CI, 0.21-1.22).⁶ It is interesting to note that the PIRs for NHL and colorectal cancer were significantly increased in the Hmong compared with all Minnesotans in 2000 through 2012, but not in 1988 through 1999 (Table 2).⁶ The PIRs for cancers of the stomach, nasopharynx, liver and cervix, were increased in Hmong versus all Minnesotans

TABLE 1. Demographics of Cancer Cases Among Hmong, Asian/Pacific Islanders, and All Minnesotans, 2000 to 2012

	Hmong	Asian/Pacific Islanders	All Minnesotans
Total no. of cases	443	3518	326,610
Average age (SD), y	53.5 (22.9)	55.8 (18.9)	64.9 (15.3)
Male, no. (%)	197 (44%)	1574 (45%)	171,232 (52%)
Female, no. (%)	246 (56%)	1944 (55%)	155,378 (48%)

Abbreviation: SD, standard deviation.

TABLE 2. PIRs of Cancer in Hmong in Minnesota, 1988 to 1999 and 2000 to 2012

Cancer Site	1988 to 1999 ^a			2000-2012		
	Observed	Expected	PIR (95% CI)	Observed	Expected	PIR (95% CI)
Nasopharynx	13	0.33	39.39 (21.01-66.86)	18	0.55	33.03 (19.60-51.95)
Stomach	21	2.41	8.70 (5.39-13.25)	31	4.28	7.24 (4.93-10.26)
Colon	11	13.9	0.79 (0.40-1.40)	NA	NA	NA
Rectum	7	5.43	1.29 (0.52-2.61)	NA	NA	NA
Colon and rectum	NA	NA	NA	50	35.63	1.40 (1.04-1.85)
Liver and intrahepatic bile duct	10	1.24	8.08 (3.88-14.71)	16	4.32	3.70 (2.12-5.98)
Pancreas	7	2.86	2.45 (0.98-4.96)	17	7.76	2.19 (1.28-3.50)
Lung and bronchus	18	18.59	0.97 (0.57-1.52)	31	41.75	0.74 (0.51-1.05)
Melanoma of the skin	2	10.69	0.19 (0.02-0.63)	12	25.58	0.47 (0.24-0.81)
Breast	11	32.98	0.33 (0.17-0.59)	24	71.12	0.33 (0.22-0.50)
Cervix uteri	14	3.76	3.72 (2.04-6.20)	29	5.06	5.73 (3.84-8.22)
Corpus and uterus, NOS	4	5.9	0.68 (0.18-1.68)	13	14.72	0.88 (0.47-1.50)
Prostate	3	19.26	0.16 (0.03-0.44)	19	48.93	0.39 (0.23-0.60)
Thyroid	7	6.1	1.15 (0.46-2.32)	19	16.58	1.15 (0.69-1.78)
Non-Hodgkin lymphoma	6	10.53	0.57 (0.21-1.22)	39	21.35	1.83 (1.30-2.50)
Acute lymphocytic leukemia	2	2.95	0.68 (0.08-2.30)	17	10	1.70 (0.99-2.71)
Acute myeloid leukemia	4	1.89	2.11 (0.57-5.24)	NA	NA	NA
Chronic myeloid leukemia	2	1.14	1.75 (0.20-5.94)	NA	NA	NA
Myeloid and monocytic leukemia	NA	NA	NA	17	7.79	2.18 (1.27-3.48)

Abbreviations: 95% CI, 95% confidence interval; NA, not available; NOS, not otherwise specified; PIR, proportional incidence ratio.

^a Data derived from Ross JA, Xie Y, Kiffmeyer WR, Bushhouse S, Robison LL. Cancer in the Minnesota Hmong population. *Cancer*. 2003;97:3076-3079.⁶

during both periods. The PIR for pancreatic cancer was significantly increased in period 2000 through 2012 but not for the period 1988 through 1999, but the number of cases was limited from 1988-1999.

The AARs between 2000 and 2012 were 185.13 per 100,000 population for Hmong individuals, 291.33 per 100,000 population for API individuals, and 472.70 per 100,000 population for all Minnesotans (Table 3). Compared with all Minnesotans, the Hmong had significantly higher AARs for cancers of the nasopharynx, stomach, liver, and cervix uteri, and significantly lower AARs for cancers of the colorectum, lung, breast, corpus uterus, and prostate, as well as for melanoma and NHL (Table 4). Compared with API individuals, the Hmong had significantly higher AARs for cancer of the cervix uteri and melanoma, and significantly lower AARs for cancers of the colorectum, liver, lung, breast, uterine corpus, and prostate (Table 4). When the 3 populations-Hmong, API, and all Minnesotans-were compared, the AAR for cancer of the cervix uteri was highest for the Hmong and lowest for all Minnesotans, whereas the AARs for cancers of the colorectum, lung, breast, uterine corpus, and prostate were in the opposite order, namely, lowest in the Hmong and highest in all Minnesotans.

DISCUSSION

The results of the current study demonstrate that, compared with all Minnesotans, the Hmong had elevated PIRs and AARs for several malignancies associated with

TABLE 3. AARs per 100,000 Population in Hmong, Asian/Pacific Islanders, and All Minnesotans, 2000 to 2012

Cancer Site	Hmong		Asian/Pacific Islanders		All Minnesotans	
	No. of Cases	AAR (95% CI)	No. of Cases	AAR (95% CI)	No. of Cases	AAR (95% CI)
All sites	443	185.13 (165.70-204.60)	3518	291.33 (281.70-300.96)	326,610	472.70 (471.00-474.30)
Nasopharynx	18	6.81 (3.38-10.25)	75	4.73 (3.55-5.91)	293	0.43 (0.38-0.48)
Stomach	31	16.71 (10.51-22.91)	143	13.77 (11.29-16.25)	3877	5.60 (5.42-5.77)
Colon and rectum	50	22.86 (15.98-29.75)	369	33.44 (29.68-37.19)	31,797	45.77 (45.27-46.28)
Liver and intrahepatic bile duct	16	7.90 (3.70-12.10)	188	16.40 (13.85-18.94)	3114	4.45 (4.30-4.61)
Pancreas	17	10.48 (5.39-15.58)	98	9.85 (7.75-11.94)	7067	10.21 (9.97-10.45)
Lung and bronchus	31	16.21 (10.31-22.12)	327	31.44 (27.77-35.11)	38,776	56.69 (56.12-57.25)
Melanoma of the skin	12	5.72 (2.36-9.07)	28	2.56 (1.52-3.61)	15,609	22.75 (22.39-23.11)
Breast	24	9.09 (5.08-13.10)	516	38.06 (34.46-41.67)	48,626	69.80 (69.18-70.42)
Cervix uteri	29	12.36 (7.49-17.23)	97	7.79 (6.08-9.51)	2142	3.19 (3.05-3.32)
Corpus and uterus, NOS	13	3.93 (1.62-6.24)	127	8.98 (7.28-10.68)	10,382	14.72 (14.44-15.01)
Prostate	19	11.27 (6.05-16.49)	255	25.82 (22.47-29.17)	54,145	78.19 (77.53-78.85)
Thyroid	19	6.84 (3.24-10.44)	204	11.84 (10.03-13.64)	6892	10.23 (9.99-10.47)
Non-Hodgkin lymphoma	39	14.66 (9.32-20.01)	192	15.02 (12.61-17.43)	15,319	22.21 (21.86-22.57)
Acute lymphocytic leukemia	17	1.87 (0.74-3.00)	57	2.34 (1.65-3.04)	991	1.50 (1.41-1.60)
Myeloid and monocytic leukemia	17	4.83 (1.76-7.89)	73	5.14 (3.71-6.58)	4415	6.43 (6.24-6.62)

Abbreviations: 95% CI, 95% confidence interval; AAR, age-adjusted cancer incidence rates; NOS, not otherwise specified.

TABLE 4. AARs per 100,000 Population for Hmong Compared to Asian/Pacific Islanders and All Minnesotans, 2000 to 2012

Cancer Site	Hmong Versus API		Hmong Versus All Minnesotans	
	Rate Ratio (95% CI)	P	Rate Ratio (95% CI)	P
All sites	0.64 (0.57-0.71)	<.0001	0.39 (0.35-0.44)	<.0001
Nasopharynx	1.44 (0.82-2.53)	.20	15.90 (9.48-26.68)	<.0001
Stomach	1.21 (0.80-1.83)	.36	2.99 (2.06-4.33)	<.0001
Colon and rectum	0.68 (0.50-0.94)	.02	0.50 (0.37-0.68)	<.0001
Liver and intrahepatic bile duct	0.48 (0.28-0.84)	.001	1.77 (1.04-3.02)	.04
Pancreas	1.06 (0.63-1.81)	.82	1.03 (0.63-1.67)	.92
Lung and bronchus	0.52 (0.35-0.76)	.001	0.29 (0.20-0.41)	<.0001
Melanoma of the skin	2.23 (1.09-4.56)	.03	0.25 (0.14-0.45)	<.0001
Breast	0.24 (0.15-0.37)	<.0001	0.13 (0.08-0.20)	<.0001
Cervix uteri	1.59 (1.01-2.49)	.05	3.88 (2.61-5.77)	<.0001
Corpus and uterus, NOS	0.44 (0.24-0.81)	.009	0.27 (0.15-0.48)	<.0001
Prostate	0.44 (0.27-0.71)	.001	0.14 (0.09-0.23)	<.0001
Thyroid	0.58 (0.33-1.00)	.05	0.67 (0.39-1.13)	.13
Non-Hodgkin lymphoma	0.98 (0.66-1.45)	.91	0.66 (0.46-0.95)	.03
Acute lymphocytic leukemia	0.80 (0.41-1.57)	.51	1.24 (0.68-2.29)	.48
Myeloid and monocytic leukemia	0.94 (0.47-1.88)	.86	0.75 (0.40-1.42)	.38

Abbreviations: 95% CI, 95% confidence interval; AAR, age-adjusted cancer incidence rates; API, Asian/Pacific Islander; NOS, not otherwise specified.

infections, including nasopharyngeal, stomach, liver, and cervical cancers. Hmong individuals also were found to have higher PIRs for NHL and myeloid leukemias. In addition, Hmong had lower AARs for all cancers combined (by 61%), as well as cancers of the colon and rectum (by 50%), lung (by 71%), breast (by 87%), uterine corpus (by 73%), and prostate (by 86%) compared with all Minnesotans. It is interesting to note that the AARs for all these cancers were lowest in the Hmong followed by API individuals, and were highest in all Minnesotans. Finally, compared with API individuals, the rates in Hmong were

significantly higher for melanoma and cervical cancer, but were lower for liver cancer.

The 4 cancers found to have increased rates in Hmong versus all Minnesotans, namely nasopharyngeal, stomach, liver, and cervical cancers, are associated with pathogens prevalent in Asia: nasopharyngeal cancer with Epstein-Barr virus (EBV), stomach cancer with *Helicobacter pylori* (*H. pylori*) bacteria, liver cancer with HBV, and cervical cancer with HPV. The relative risks for nasopharyngeal and stomach cancers in the Hmong were similar to those in API individuals. However, for Hmong, the

rate of cervical cancer was 60% higher and the rate of liver cancer was 52% lower than among APIs, and the risk of cervical cancer was highest in the Hmong and the risk of liver cancer was highest in API individuals.

Among these infection-related cancers, the highest AAR ratio for the Hmong versus all Minnesotans was observed for nasopharyngeal cancer. AAR rates were 15.90 times higher and the PIRs were 33 times higher in Hmong compared with all Minnesotans. These findings are similar to the findings in a prior study by Ross et al.⁶ In addition to a higher prevalence of EBV in the Hmong, the higher risk of nasopharyngeal cancer in this population may be attributed to differences in genetics between Hmong and other Minnesota residents. For example, Hmong had higher frequencies of glutathione S-transferase U1 and glutathione S-transferase O1 null genotypes compared with white Minnesotans. These null genotypes have been linked to nasopharyngeal cancer.¹¹

High rates of cervical cancer for the Hmong and API individuals may be explained by the higher prevalence of HPV infection because it is endemic in Asia and because of lower HPV vaccination rates among young Asian Americans. In a survey conducted among undergraduate students in a public university in the Midwest between 2012 and 2014, the HPV vaccination rate was 60.7% for non-Latina women and 38.6% for API women.¹² A recent study in Minnesota also reported that the initiation rate of vaccination was 46.3% and the completion rate was 32.7% among Hmong American immigrants,¹³ whereas the rate for the Papanicolaou test was 67.1% within the last 3 years.¹⁴ In keeping with these findings, approximately 74% of Hmong women in California had ever undergone a Papanicolaou test and approximately 61% had been screened within the last 3 years compared with rates of 91% and 86%, respectively, in women in California overall.¹⁵

Similarly, the high age-adjusted rates of hepatocellular carcinoma may be explained by the high prevalence of HBV infection among Hmong immigrants. A study in California reported that only 12% of Hmong youth completed the 3-shot HBV vaccination series compared with a completion rate of 78.8% observed for seventh graders of all ethnicities.¹⁶ A recent study performed in Oregon demonstrated that approximately 50.6% of Hmong Americans were vaccinated for HBV.¹⁷ Thus, there is a need for well-designed intervention measures that will improve the vaccination rates among Hmong individuals. A randomized controlled trial, a lay health worker model in California, demonstrated that an intervention focused on modifying individual health behavior framework

factors may significantly improve HBV screening rates. The intervention group was found to have a significantly higher likelihood of reporting HBV serology testing (24%) compared with the control group (10%).¹⁸

The higher risk of stomach cancer observed in the Hmong versus all Minnesotans may be partially explained by a high prevalence of *H. pylori* infection, which is a strong risk factor for gastric cancers.^{19,20} A study in Thailand demonstrated a high prevalence of metronidazole-resistant *H. pylori* strains among the Hmong population compared with the Thai and Karen populations.²¹ To the best of our knowledge, there currently are no data regarding the prevalence of *H. pylori* infection in the Hmong in the United States. In addition to *H. pylori*, a high consumption of rice may explain the high incidence of stomach cancer observed among the Hmong.²²

In the PIR analysis, a higher incidence of NHL and myeloid leukemias was observed in the Hmong compared with all Minnesotans. This finding is in parallel to findings for Hmong in California.⁷ The higher incidence of NHL might be explained by a higher prevalence of EBV infection in Hmong, which is associated with the risk of NHL.^{23,24}

A systematic review identified barriers to cancer screening in the Hmong, including access to health care, cultural norms, and cancer literacy.²⁵ Results from a focus group demonstrated that Hmong women have sociocultural norms that discourage them from seeking screening,²⁶ have a strong adherence to traditional medicine,²⁷ and do not actively seek help.²⁸ In addition, Hmong leaders in Wisconsin indicated that the Hmong believed that treating cancer must be done by a spiritual healer rather than a physician, and expressed concern regarding the lack of outreach regarding different cancers to their communities.²⁹

The lower rate of lung cancer in the Hmong may be explained by the low smoking prevalence in this population: approximately 7% of Hmong respondents in Wisconsin reported ever smoking >100 cigarettes in their entire lifetime.²⁹ In the 2007 Minnesota Adult Tobacco Survey,^{30,31} approximately 5% of Hmong adults in Minnesota reported smoking compared with 17% of all Minnesotans. Currently, smoking rates are on the rise in the acculturated Hmong youth,³² whereas they are decreasing among all Minnesotans.³³ This can lead to an increase in lung cancer rates in the Hmong population in the future.

Furthermore, the lower rates of breast, colorectal, and prostate cancers noted in Hmong individuals versus all Minnesotans may be explained in part by the lower prevalence of underlying risk factors for these cancers,

such as obesity and smoking, in the Hmong, and/or the healthy immigrant effect. Studies have demonstrated a pattern of initial good health in immigrants compared with their US-born adult counterparts.³⁴ In addition, a lower incidence of breast, colorectal, and prostate cancers in the Hmong could be attributed to the low screening rates noted in this population. Without screening, these cancers may be undiagnosed and therefore not reported. In agreement with this explanation, it was shown that cancers are diagnosed in Hmong individuals at a later stage compared with API and non-Hispanic white individuals.⁷ The low rate of breast cancer also might be due to the high parity and low age at first pregnancy among Hmong women. Although these rates currently are low, it is important to continue the surveillance of these cancers in the Hmong.

Acculturation in the Hmong has led to increases in weight and changes in dietary habits resulting in the low consumption of fruits and vegetables,³⁵ which could lead to increased cancer rates in the future.³⁶ A higher risk of breast cancer has been observed in Asian American women who have resided in the United States for >10 years compared with women who recently immigrated.³⁷

Compared with API individuals, lower rates of colon, prostate, liver, and thyroid cancer, and higher rates of cervical cancer and melanoma, were noted among Hmong individuals. These higher rates of melanoma in the Hmong compared with APIs might be related to their occupation, because a large percentage of Hmong in the United States engage in agricultural jobs, which is a largely outdoor occupation.⁷ Thus, there should be better education regarding preventive measures (such as applying sunscreen) in Hmong in Minnesota.

To our knowledge, the current study is the first to estimate AARs in the Hmong individuals residing in Minnesota compared with all Minnesotans. Furthermore, we also used PIRs to identify the relative percentage of specific cancer counts in Hmong and compared them with a previous study performed in 1998 through 1999. Although the previous study identified Hmong cases through surname matching, the current study used race as recorded in the medical record.⁶ We did not use surname matching mostly because the reporting of race information was much more complete in the latter time period for the MCSS. In addition, Chinese and other non-Hmong Minnesota residents have last names similar to Hmong, which could lead to an overestimation of cancer cases in Hmong and increased rates.^{7,38}

A limitation of the current study is the potential for an undercounting of Hmong cancer cases in the MCSS.

Race classification in the MCSS is based on what is reported by facilities and their medical records. This may or may not be accurate depending on how precise facilities are at identifying their patients. In addition, the reports from the US Census are likely to slightly underestimate the population counts of Hmong in Minnesota by approximately 2%.³⁹ Another limitation is that the number of Hmong cancer cases in Minnesota diagnosed between 2000 and 2012 was small (443 cases), which led to wide 95% CIs, thereby making it more difficult to observe any disparities that may exist. In addition, residual confounding by age could contribute to a smaller percentage of cancer cases in Hmong individuals or inaccurate denominators may be partially responsible for discrepancies in cancer rates between Hmong and API individuals.

The current study brings awareness of the higher risk of cancers of the nasopharynx, stomach, liver, and cervix uteri among the Hmong, cancers that are associated with different infections. In the future, we expect that the incidence of cancers caused by infectious agents will decrease among successive generations in the children and grandchildren of immigrants as acculturation occurs and vaccine use becomes more commonly accepted. In the interim, the Hmong should follow screening recommendations for cancer in the general population because it has been reported that Hmong individuals tend to use cancer screening services less often than non-Hispanic white individuals. Efforts to expand preventative approaches for cancer in the Minnesota Hmong population are warranted.

FUNDING SUPPORT

The collection of the cancer incidence data used in this publication was partially supported by Cooperative Agreement Number 5 NU58DP003922 from the Centers for Disease Control and Prevention. Its contents are solely the responsibility of the authors and do not necessarily represent the official views of the Centers for Disease Control and Prevention.

CONFLICT OF INTEREST DISCLOSURES

The authors made no disclosures.

AUTHOR CONTRIBUTIONS

Mosunmoluwa Oyenuga: Writing-original draft and writing-review and editing. **John K. Yang:** Conceptualization, formal analysis, and writing-original draft. **Anna E. Prizment:** Methodology, conceptualization, and writing-review and editing. **Sally Bush-house:** Formal analysis and writing-review and editing. **Ellen Demerath:** Writing-review and editing. **Logan G. Spector:** Supervision and writing-review and editing.

REFERENCES

- Minnesota Department of Health. Refugee health statistics; cumulative arrivals, 1979-2014. <http://www.health.state.mn.us/divs/idepc/refugee/index.html>. Accessed June 24, 2017.
- Pfeifer ME, Sullivan J, Yang K, Yang W. Hmong population and demographic trends in the 2010 Census and 2010 American Community Survey. *Hmong Studies J*. 2012;13:2010 Census Special Issue.
- US Census Bureau. Census 2000 summary file 2 (SF 2) 100-percent data; table DP-1. <http://factfinder2.census.gov>. Accessed June 24, 2017.
- US Census Bureau. 2010 census summary file 2 (SF 2); table DP-1. <http://factfinder2.census.gov>. Accessed June 24, 2017.
- Pinzon-Perez H. Health issues for the Hmong population in the U.S.: implications for health educators. *Int Electr J Health Educ*. 2006;9:122-133.
- Ross JA, Xie Y, Kiffmeyer WR, Bushhouse S, Robison LL. Cancer in the Minnesota Hmong population. *Cancer*. 2003;97:3076-3079.
- Mills PK, Yang RC, Riordan D. Cancer incidence in the Hmong in California, 1988-2000. *Cancer*. 2005;104(suppl 12):2969-2974.
- Minnesota Cancer Surveillance System. Cancer in Minnesota, 1988-2008. St. Paul, MN: Minnesota Department of Health; 2012.
- Ury HK, Wiggins AD. Another shortcut method for calculating the confidence interval of a Poisson variable (or of a standardized mortality ratio). *Am J Epidemiol*. 1985;122:197-198.
- Fleiss JL, Levin B, Paik MC. Statistical Methods for Rates and Proportions. 3rd ed. Hoboken, NJ: John Wiley & Sons; 2003.
- Kiffmeyer WR, Langer E, Davies SM, Envall J, Robison LL, Ross JA. Genetic polymorphisms in the Hmong population: implications for cancer etiology and survival. *Cancer*. 2004;100:411-417.
- Lee HY, Kwon M, Vang S, et al. Disparities in human papillomavirus vaccine literacy and vaccine completion among Asian American Pacific Islander undergraduates: implications for cancer health equity. *J Am Coll Health*. 2015;63:316-323.
- Beltran R, Simms T, Lee HY, Kwon M. HPV literacy and associated factors among Hmong American immigrants: implications for reducing cervical cancer disparity. *J Community Health*. 2016;41:603-611.
- Lee HY, Yang PN, Lee DK, Ghebre R. Cervical cancer screening behavior among Hmong-American immigrant women. *Am J Health Behav*. 2015;39:301-307.
- Fang DM, Lee S, Stewart S, Ly MY, Chen MS Jr. Factors associated with Pap testing among Hmong women. *J Health Care Poor Underserved*. 2010;21:839-850.
- Burdler LM, Mills PK, Yang RC, Chen MS Jr. Hepatitis B knowledge and vaccination levels in California Hmong youth: implications for liver cancer prevention strategies. *Asian Pac J Cancer Prev*. 2005;6:401-403.
- Kue J, Thorburn S. Hepatitis B knowledge, screening, and vaccination among Hmong Americans. *J Health Care Poor Underserved*. 2013;24:566-578.
- Chen MS Jr, Fang DM, Stewart SL, et al. Increasing hepatitis B screening for Hmong adults: results from a randomized controlled community-based study. *Cancer Epidemiol Biomarkers Prev*. 2013;22:782-791.
- Parsonnet J, Friedman GD, Vandersteen DP, et al. Helicobacter pylori infection and the risk of gastric carcinoma. *N Engl J Med*. 1991;325:1127-1131.
- Correa P, Piazuelo MB. Helicobacter pylori infection and gastric adenocarcinoma. *US Gastroenterol Hepatol Rev*. 2011;7:59-64.
- Vilaichone R, Ratanachu-Ek T, Gamnarai P, et al. Extremely high prevalence of metronidazole-resistant helicobacter pylori strains in mountain people (Karen and Hmong) in Thailand. *Am J Trop Med Hyg*. 2016;94:717-720.
- Yang RC, Mills PK, Riordan DG. Gastric adenocarcinoma among Hmong in California, USA, 1988-2000. *Gastric Cancer*. 2005;8:117-123.
- Heslop HE. Biology and treatment of Epstein-Barr virus-associated non-Hodgkin lymphomas. *Hematology Am Soc Hematol Educ Program*. 2005;2005:260-266.
- Teras LR, Rollison DE, Pawlita M, et al. Epstein-Barr virus and risk of non-Hodgkin lymphoma in the Cancer Prevention Study-II and a meta-analysis of serologic studies. *Int J Cancer*. 2014;136:108-116.
- Lee HY, Vang S. Barriers to cancer screening in Hmong Americans: the influence of health care accessibility, culture, and cancer literacy. *J Community Health*. 2010;35:302-314.
- Lor M, Khang PY, Xiong P, Moua KF, Lauver D. Understanding Hmong women's beliefs, feelings, norms, and external conditions about breast and cervical cancer screening. *Public Health Nurs*. 2013;30:420-428.
- Baisch MJ, Vang PC, Peterman BR. An exploration of Hmong women's perspectives on cancer. *Asian Nurs Res (Korean Soc Nurs Sci)*. 2008;2:82-91.
- Smalkoski K, Herther NK, Xiong ZB, Ritsema K, Vang R, Zheng R. Health disparities research in the Hmong American community: implications for practice and policy. *Hmong Studies J*. 2012;13:2010 Census Special Issue.
- Schroepfer TA, Waltz A, Noh H, Matloub J, Kue V. Seeking to bridge two cultures: the Wisconsin Hmong cancer experience. *J Cancer Educ*. 2010;25:609-616.
- ClearWay Minnesota, Blue Cross and Blue Shield of Minnesota, and Minnesota Department of Health. Creating a healthier Minnesota: progress in reducing tobacco use. September 2008. Minneapolis, MN. <http://www.health.state.mn.us/divs/chs/tobacco/matscomptech07.pdf>. Accessed May 29, 2018.
- Blue Cross and Blue Shield of Minnesota, ClearWay Minnesota, Asian Pacific Tobacco-Free Coalition of Minnesota, Southeast Asian Refugee Community Home. Tobacco use in Minnesota: a quantitative survey of Cambodian, Hmong, Lao and Vietnamese community members. November 2009. <http://clearwaymn.org/wp-content/uploads/2012/12/Tabacco-Use-in-Minnesota-2010-Full-Report.pdf>. Accessed May 29, 2018.
- Burgess DJ, Mock J, Schillo BA, et al. Culture, acculturation and smoking use in Hmong, Khmer, Laotians, and Vietnamese communities in Minnesota. *BMC Public Health*. 2014;14:791.
- ClearWay Minnesota and Minnesota Department of Health. Tobacco use in Minnesota: 2010 update. February 2011. Minneapolis, MN. <http://www.health.state.mn.us/divs/chs/tobacco/matscomptech07.pdf>. Accessed May 29, 2018.
- Hamilton TG. The healthy immigrant (migrant) effect: in search of a better native-born comparison group. *Soc Sci Res*. 2015;54:353-365.
- Yang RC, Mills PK. Dietary and lifestyle practices of Hmong in California. *J Health Care Poor Underserved*. 2008;19:1258-1269.
- Franzen L, Smith C. Acculturation and environmental change impacts dietary habits among adult Hmong. *Appetite*. 2009;52:173-183.
- Chen MS Jr. Cancer health disparities among Asian Americans: what we do and what we need to do. *Cancer*. 2005;104(suppl 12):2895-2902.
- Mills PK, Yang R. Cancer incidence in the Hmong of Central California, United States, 1987-94. *Cancer Causes Control*. 1997;8:705-712.
- US Census Bureau. Census Bureau releases estimates of undercount and overcount in the 2010 Census. https://www.census.gov/newsroom/releases/archives/2010_census/cb12-95.html. Accessed April 10, 2018.